

DEPARTMENT : “C.S.E.”

SEMESTER : 1

ASSIGNMENT TITLE:

How Can You Demonstrate Linux Setup,
Command Usage, and Automation Through
Practical Implementation?

COURSE NAME:

Computer Science and Fundamentals &
career pathway

COURS CODE: ETCCCP105

FACULTY

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SECTION: “B”

2.1) Linux Installation

Introduction

I installed *Ubuntu Linux* on my Windows laptop using *Windows Subsystem for Linux (WSL 2)*.

WSL 2 provides a real Linux environment inside Windows without requiring a separate virtual machine.

This setup makes it simple and safe to practice Linux commands while still using Windows as the main operating system.

Step 1: Enabling WSL

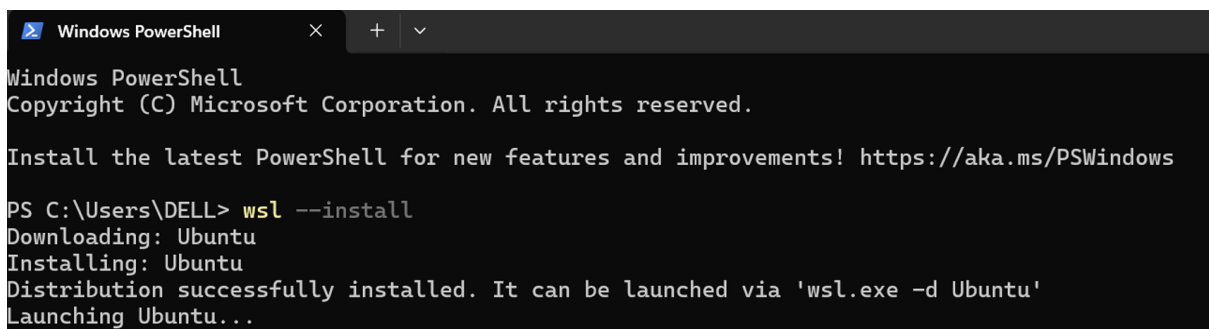
I first opened the *Run* dialog using *Windows + R*, typed *powershell*, and chose *Run as Administrator*.

Inside PowerShell, I executed the command that enables:

Windows Subsystem for Linux

Virtual Machine Platform

Once the installation is completed, I restarted my system.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\DELL> wsl --install
Downloading: Ubuntu
Installing: Ubuntu
Distribution successfully installed. It can be launched via 'wsl.exe -d Ubuntu'
Launching Ubuntu...
```

Step 2: Installing Ubuntu

1. I opened the *Microsoft Store*
2. Searched for *Ubuntu 22.04 LTS*
3. Clicked *Install*
4. After the download finished, I launched it

Step 3: First-Time Ubuntu Setup

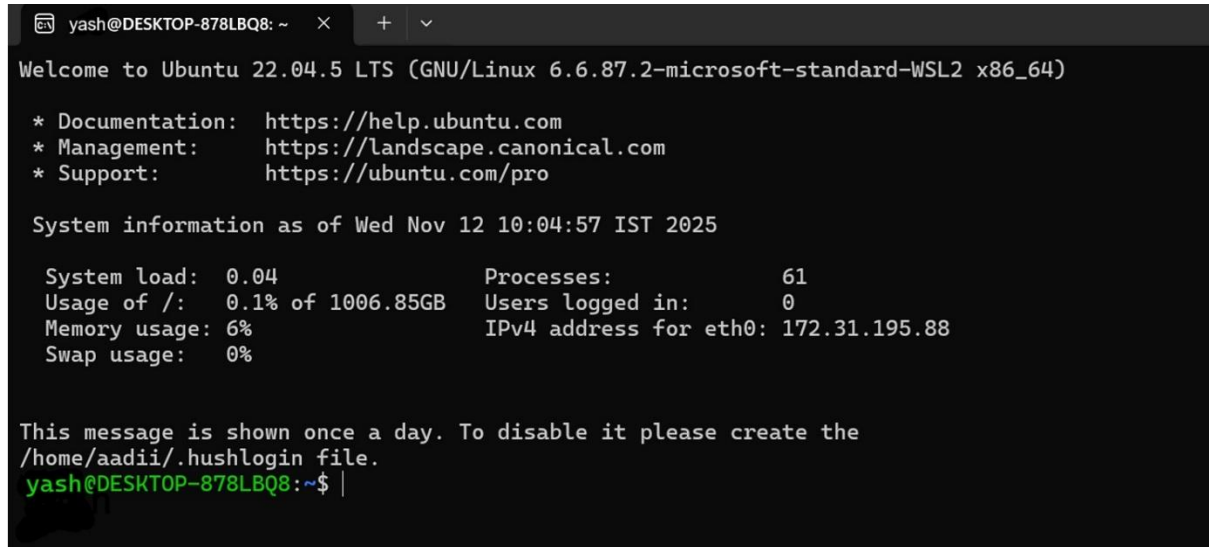
The first launch shows a short installation message.

It then asks for:

A username

A password

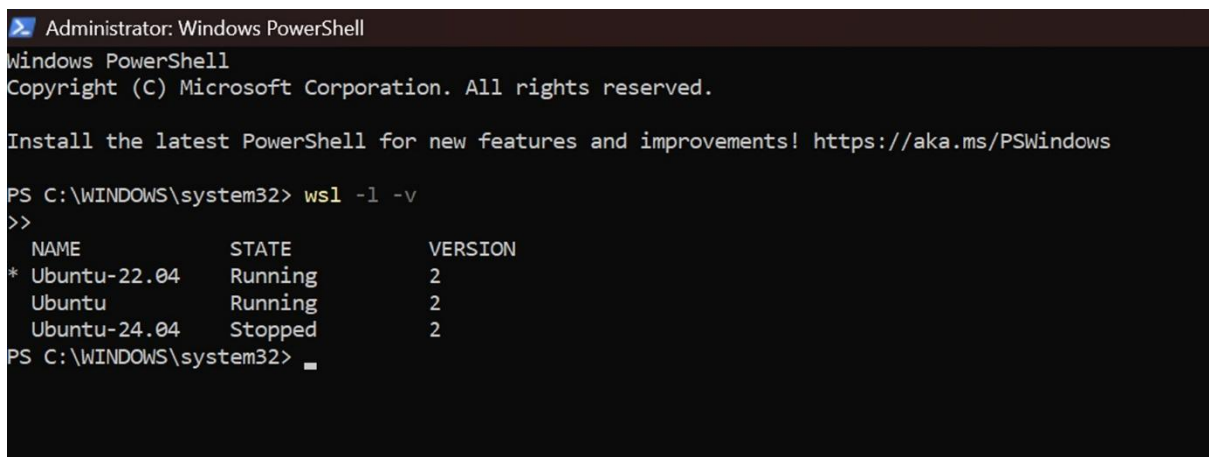
After setting these, the Ubuntu terminal becomes ready to use.



```
yash@DESKTOP-878LBQ8: ~  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Wed Nov 12 10:04:57 IST 2025  
  
System load:  0.04          Processes:            61  
Usage of /:   0.1% of 1006.85GB Users logged in:       0  
Memory usage: 6%           IPv4 address for eth0: 172.31.195.88  
Swap usage:   0%  
  
This message is shown once a day. To disable it please create the  
/home/aadii/.hushlogin file.  
yash@DESKTOP-878LBQ8:~$ |
```

Step 4: Verifying Installation

To ensure that Ubuntu was correctly installed, I ran a simple verification command in the terminal, which confirmed that WSL 2 was functioning properly.



```
Administrator: Windows PowerShell  
Windows PowerShell  
Copyright (C) Microsoft Corporation. All rights reserved.  
  
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows  
  
PS C:\WINDOWS\system32> wsl -l -v  
>>  
NAME                STATE              VERSION  
* Ubuntu-22.04      Running            2  
  Ubuntu            Running            2  
  Ubuntu-24.04      Stopped            2  
PS C:\WINDOWS\system32> █
```

Step 5: Final Working Environment

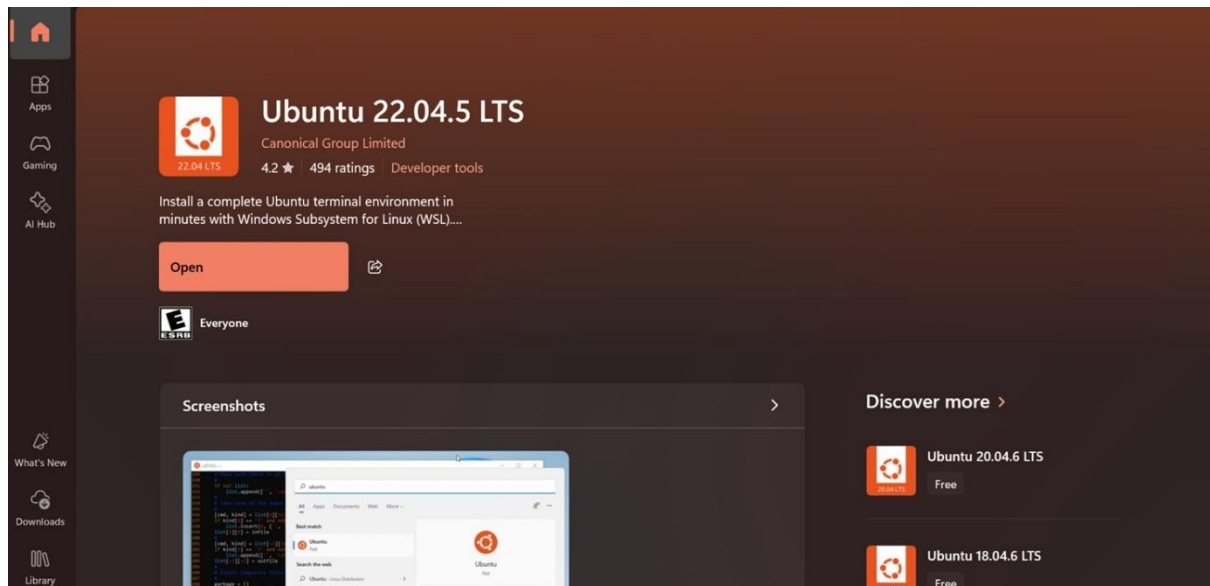
Ubuntu now runs directly inside Windows as a fully functional Linux command-line environment.

I can run Linux commands, write shell scripts, and perform all tasks without needing any virtual machine.

Hardware Configuration of My System

- **System Manufacturer:** Dell Inc.
- **Model:** Latitude 5420
- **Processor:** 11th Gen Intel® Core™ i5-1145G7 @ 2.60GHz (8 CPUs)
- **Installed RAM:** 16 GB
- **System Type:** 64-bit, x64-based processor
- **Graphics:** Intel® Iris® Xe Graphics
- **Storage:** 238 GB SSD
- **WSL Disk Allocation:** Handled dynamically by Windows
- **Virtualization:** WSL 2 using Hyper-V backend

Ubuntu 22.04 LTS is now completely functional inside my Windows system and ready for Linux-based practical work.



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> wsl -l -v
>>
  NAME                STATE      VERSION
* Ubuntu-22.04        Running    2
  Ubuntu               Running    2
  Ubuntu-24.04         Stopped    2
PS C:\WINDOWS\system32>
```

```
yash@DESKTOP-878LBQ8: ~  ×  +  ▾
yash@DESKTOP-878LBQ8:~$ |
```

2.2) Shell Commands: Implementation & Documentation

Below are the commands I practiced, along with their purpose and how I used them.

1) *pwd*

Purpose: Shows the directory in which I am currently working.

Usage: Verified that I was inside my home directory */home/yash*.

```
yash@DESKTOP-878LBQ8: ~  ×  +  ▾
yash@DESKTOP-878LBQ8:~$ cd ~
yash@DESKTOP-878LBQ8:~$ pwd
/home/yash
yash@DESKTOP-878LBQ8:~$ |
```

2) *ls*

Purpose: Lists files and folders inside the current directory.

Usage: Used it to check items inside the home directory and inside *testfolder*.

```
yash@DESKTOP-878LBQ8:~$ ls
snap  testfolder
yash@DESKTOP-878LBQ8:~$ |
```

3) *cd*

Purpose: Used to move between directories.

Usage:

- *cd ~* to return to the home directory
- *cd testfolder* to enter the folder I created

```
yash@DESKTOP-878LBQ8:~$ mkdir testfolder
cd testfolder
mkdir f1 f2
touch file1.txt file2.txt
mkdir: cannot create directory 'testfolder': File exists
mkdir: cannot create directory 'f1': File exists
mkdir: cannot create directory 'f2': File exists
yash@DESKTOP-878LBQ8:~/testfolder$ |
```

4) *tree*

Purpose: Displays files and folders in a tree-style format.

Usage: After creating folders (f1, f2) and some files, I used ``tree`` to view the structure.

```
yash@DESKTOP-878LBQ8:~/testfolder$ tree
.
├── cat file1.txt
├── demo
│   └── file1.txt
├── f1
├── f2
├── file1.txt
├── file2.txt
└── renamed.txt

4 directories, 5 files
yash@DESKTOP-878LBQ8:~/testfolder$ |
```

5) *mkdir*

Purpose: Creates a new directory.

Usage: Created a folder named *demo* inside *testfolder* using *mkdir demo*.

```
yash@DESKTOP-878LBQ8:~/testfolder$ mkdir demo
mkdir: cannot create directory 'demo': File exists
yash@DESKTOP-878LBQ8:~/testfolder$ ls
'cat file1.txt'  demo  f1  f2  file1.txt  file2.txt  renamed.txt
```

6) *touch*

Purpose: Creates an empty file.

Usage: Used to create *newfile.txt* for testing.

```
yash@DESKTOP-878LBQ8:~/testfolder$ touch newfile.txt
ls
'cat file1.txt'  demo  f1  f2  file1.txt  file2.txt  newfile.txt  renamed.txt
```

7) *cp*

Purpose: Copies files or directories.

Usage: Copied *file1.txt* into the *demo* folder.

```
yash@DESKTOP-878LBQ8:~/testfolder$ cp file1.txt demo/
ls demo
file1.txt
```

8) *mv*

Purpose: Moves or renames a file.

Usage: Renamed *file2.txt* to *renamed.txt*.

```
yash@DESKTOP-878LBQ8:~/testfolder$ mv file2.txt renamed.txt
ls
'cat file1.txt'  demo  f1  f2  file1.txt  newfile.txt  renamed.txt
```

9) *rm*

Purpose: Deletes files.

Usage: Removed *newfile.txt* after testing.

```
yash@DESKTOP-878LBQ8:~/testfolder$ rm newfile.txt
ls
'cat file1.txt'  demo  f1  f2  file1.txt  renamed.txt
yash@DESKTOP-878LBQ8:~/testfolder$
```

10) *chmod*

Purpose: Changes file permissions.

Usage: Gave full permissions (read, write, execute) to *file1.txt* using *chmod 777*.

```

yash@DESKTOP-878LBQ8:~/testfolder$ ls -l
total 20
-rw-r--r-- 1 yash yash 22 Nov 12 18:33 'cat file1.txt'
drwxr-xr-x 2 yash yash 4096 Nov 12 18:21 demo
drwxr-xr-x 2 yash yash 4096 Nov 12 18:11 f1
drwxr-xr-x 2 yash yash 4096 Nov 12 18:11 f2
-rwxrwxrwx 1 root yash 21 Nov 13 10:26 file1.txt
-rw-r--r-- 1 yash yash 0 Nov 13 10:26 renamed.txt
yash@DESKTOP-878LBQ8:~/testfolder$ chmod 777 file1.txt
chmod: changing permissions of 'file1.txt': Operation not permitted
yash@DESKTOP-878LBQ8:~/testfolder$ ls -l
total 20
-rw-r--r-- 1 yash yash 22 Nov 12 18:33 'cat file1.txt'
drwxr-xr-x 2 yash yash 4096 Nov 12 18:21 demo
drwxr-xr-x 2 yash yash 4096 Nov 12 18:11 f1
drwxr-xr-x 2 yash yash 4096 Nov 12 18:11 f2
-rwxrwxrwx 1 root yash 21 Nov 13 10:26 file1.txt
-rw-r--r-- 1 yash yash 0 Nov 13 10:26 renamed.txt

```

11) chown

Purpose: Changes file ownership.

Usage: Changed the owner of *file1.txt* from my user to root.

```

yash@DESKTOP-878LBQ8:~/testfolder$ sudo chown root file1.txt
[sudo] password for aadii:
yash@DESKTOP-878LBQ8:~/testfolder$ ls -l
total 20
-rw-r--r-- 1 yash yash 22 Nov 9 18:33 'cat file1.txt'
drwxr-xr-x 2 yash yash 4096 Nov 9 18:21 demo
drwxr-xr-x 2 yash yash 4096 Nov 9 18:11 f1
drwxr-xr-x 2 yash yash 4096 Nov 9 18:11 f2
-rwxrwxrwx 1 root yash 21 Nov 12 10:26 file1.txt
-rw-r--r-- 1 yash yash 0 Nov 12 10:26 renamed.txt

```

12) ps

Purpose: Shows running processes.

Usage: Used it to view processes active under my Ubuntu session.

```

yash@DESKTOP-878LBQ8:~/testfolder$ ps
  PID TTY          TIME CMD
  476 pts/0        00:00:00 bash
  776 pts/0        00:00:00 ps

```

13) top

Purpose: Displays real-time system resource usage.

Usage: Observed CPU and RAM usage.

```
top - 11:36:47 up 1:23, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 27 total, 1 running, 26 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7634.2 total, 6995.3 free, 384.3 used, 254.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7099.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
83	root	20	0	23104	5760	4480	S	0.3	0.1	0:01.62	systemd-udevd
217	root	20	0	2219304	35548	23168	S	0.3	0.5	0:01.29	snappd
1	root	20	0	165832	10496	7936	S	0.0	0.1	0:01.38	systemd
2	root	20	0	3072	1792	1792	S	0.0	0.0	0:00.01	init-systemd(Ub
6	root	20	0	3088	1920	1920	S	0.0	0.0	0:00.00	init
60	root	19	-1	47812	14464	13824	S	0.0	0.2	0:00.39	systemd-journal
94	root	20	0	377428	6712	1408	S	0.0	0.1	0:00.12	snappfuse
97	root	20	0	526824	11868	1280	S	0.0	0.2	0:02.91	snappfuse
100	root	20	0	153136	1796	1280	S	0.0	0.0	0:00.00	snappfuse
148	systemd+	20	0	25672	13568	9088	S	0.0	0.2	0:00.13	systemd-resolve
151	systemd+	20	0	89364	7168	6400	S	0.0	0.1	0:00.34	systemd-timesyn
207	root	20	0	4308	2560	2432	S	0.0	0.0	0:00.01	cron
209	message+	20	0	8584	4224	3840	S	0.0	0.1	0:00.22	dbus-daemon
214	root	20	0	30136	18432	9984	S	0.0	0.2	0:00.09	networkd-dispat
215	syslog	20	0	222404	4992	4224	S	0.0	0.1	0:00.06	rsyslogd
218	root	20	0	15320	7296	6528	S	0.0	0.1	0:00.14	systemd-logind
235	root	20	0	3240	2176	2048	S	0.0	0.0	0:00.00	agetty
246	root	20	0	107164	21504	13440	S	0.0	0.3	0:00.06	unattended-upgr
361	root	20	0	7532	4736	3968	S	0.0	0.1	0:00.01	login
405	aadii	20	0	6104	4992	3328	S	0.0	0.1	0:00.01	bash
464	root	20	0	3196	2176	2048	S	0.0	0.0	0:00.00	agetty
469	root	20	0	3080	900	768	S	0.0	0.0	0:00.00	SessionLeader
470	root	20	0	3096	1032	896	S	0.0	0.0	0:00.02	Relay(476)
476	aadii	20	0	6128	4992	3328	S	0.0	0.1	0:00.08	bash
781	aadii	20	0	7808	3584	3072	R	0.0	0.0	0:00.02	top
784	root	20	0	23104	3416	1920	S	0.0	0.0	0:00.00	systemd-udevd

14) kill

Purpose: Terminates processes.

Usage: Stopped a background process created using *sleep*.

```
yash@DESKTOP-878LBQ8:~/testfolder$ sleep 1000 &
[1] 802
yash@DESKTOP-878LBQ8:~/testfolder$ [1] 1234
[1]: command not found
yash@DESKTOP-878LBQ8:~/testfolder$ kill 1234
-bash: kill: (1234) - No such process
yash@DESKTOP-878LBQ8:~/testfolder$ ps
  PID TTY          TIME CMD
  476 pts/0        00:00:00 bash
   802 pts/0        00:00:00 sleep
   821 pts/0        00:00:00 ps
```

15) ping

Purpose: Checks network connectivity.

Usage: Tested internet connection by pinging Google.

```
yash@DESKTOP-878LBQ8:~/testfolder$ ping -c 4 google.com
PING google.com (142.251.220.46) 56(84) bytes of data.
64 bytes from hkg07s50-in-f14.1e100.net (142.251.220.46): icmp_seq=1 ttl=110 time=75.6 ms
64 bytes from pnbomb-ba-in-f14.1e100.net (142.251.220.46): icmp_seq=2 ttl=110 time=87.3 ms
64 bytes from pnbomb-ba-in-f14.1e100.net (142.251.220.46): icmp_seq=3 ttl=110 time=75.9 ms
64 bytes from pnbomb-ba-in-f14.1e100.net (142.251.220.46): icmp_seq=4 ttl=110 time=71.9 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 71.935/77.681/87.315/5.775 ms
```

16) ip a

Purpose: Shows network interface details and IP addresses.

Usage: Checked the WSL assigned IP.

```
yash@DESKTOP-878LBQ8:~/testfolder$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:0b:26:7c brd ff:ff:ff:ff:ff:ff
    inet 172.31.195.88/20 brd 172.31.207.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe0b:267c/64 scope link
        valid_lft forever preferred_lft forever
```

17) netstat -tulnp

Purpose: Displays active listening ports.

Usage: Checked which services were using ports.

```
yash@DESKTOP-878LBQ8:~/testfolder$ sudo apt install net-tools
[sudo] password for yash:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
net-tools is already the newest version (1.60+git20181103.0eebece-1ubuntu5).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
yash@DESKTOP-878LBQ8:~/testfolder$ netstat -tulnp
(No info could be read for "-p": geteuid()=1000 but you should be root.)
Active Internet connections (only servers)

```

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State	PID/Program name
tcp	0	0	10.255.255.254:53	0.0.0.0:*	LISTEN	-
udp	0	0	10.255.255.254:53	0.0.0.0:*	-	-
udp	0	0	127.0.0.1:323	0.0.0.0:*	-	-
udp6	0	0	:::1:323	:::*	-	-

18) echo

Purpose: Prints text on the terminal.

Usage: Used to display a message as output.

```
yash@DESKTOP-878LBQ8:~/testfolder$ echo "Hello Linux"
Hello Linux
```

19) cat

Purpose: Shows the contents of files.

Usage: Viewed the content of file1.txt after editing.

```
yash@DESKTOP-878LBQ8:~/testfolder$ cat file1.txt
This is a test file.
```

20) nano

Purpose: A terminal-based text editor.

Usage: Edited text inside *file1.txt* using nano.

```
yash@DESKTOP-878LBQ8:~/testfolder$ cat file1.txt
This is a test file.
```

2.3) Shell Script Development

Script 1:backup.sh

This script creates a backup of a selected directory and adds a timestamp to the backup folder.

```
yash@DESKTOP-878LBQ8:~/testfolder$ #!/bin/bash
# Script Name: backup.sh
# Author: Yash Mehta
# Date: $(date)
# Description: Creates a backup copy of a directory with timestamp.

src="/home/yash/testfolder"
dest="/home/yash/backup_$(date +%Y%m%d_%H%M%S)"

mkdir -p "$dest"
cp -r "$src" "$dest"

echo "Backup created successfully at $dest"
#!/bin/bash
# Script Name: backup.sh
# Author: Yash Mehta
# Date: $(date)
# Description: Creates a backup copy of a directory with timestamp.

src="/home/yash/testfolder"
dest="/home/yash/backup_$(date +%Y%m%d_%H%M%S)"

mkdir -p "$dest"
cp -r "$src" "$dest"

echo "Backup created successfully at $dest"
[1]+  Done                  sleep 1000
Backup created successfully at /home/yash/backup_20251112_202220
Backup created successfully at /home/yash/backup_20251112_202220
```

Script 2: monitor.sh

This script keeps track of CPU and RAM usage and logs the data after every few seconds.

```
GNU nano 6.2 monitor.sh *
#!/bin/bash
# Script Name: monitor.sh
# Author:Yash Mehta
# Date: $(date)
# Description: Logs CPU and RAM usage to a file every 5 seconds.

while true
do
    echo "----- $(date) -----" >> system_log.txt
    top -b -n1 | head -5 >> system_log.txt
    sleep 5
done
```

Script 3: download.sh

This script automatically downloads a file from the internet and saves it inside the Downloads folder.

```
GNU nano 6.2 download.sh *
#!/bin/bash
# Script Name: download.sh
# Author:Yash Mehta
# Date: $(date)
# Description: Automatically downloads a file from the internet.

url="https://example.com/file.txt"
dest="/home/yash/Downloads"

mkdir -p "$dest"
wget -P "$dest" "$url"

echo "File downloaded successfully to $dest"
```

```
yash@DESKTOP-878LBQ8:~$ nano download.sh
yash@DESKTOP-878LBQ8:~$ chmod +x download.sh
./download.sh
--2025-11-12 20:30:23-- https://example.com/file.txt
Resolving example.com (example.com)... 23.220.75.245, 23.192.228.80, 23.215.0.136, ...
Connecting to example.com (example.com)|23.220.75.245|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-11-12 20:30:24 ERROR 404: Not Found.
```

GitHub Repository

<https://github.com/yashmehta53-cyk/Fundamentals-Assignment-2>